

U.S. 4-year College Institution Graduation Rates

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Introduction | Literature Review | Project Description

Colleges in the United States face two big challenges: making sure more students can enroll *and* helping those who do enroll actually finish their degrees. We often look at the graduation rate – *the amount of first-time, full-time students who earn a bachelor's degree within six years* – to see how well a college is doing, (National Center for Education Statistics [NCES], 2023; U.S. Department of Education, n.d.). At the same time, college institutions differ in how selective they are (admission rate), how much they charge (cost of attendance), and how many students receive Pell Grants. By exploring how these factors relate to graduation rates, educators and policymakers can build better support programs, set fair tuition levels, and target financial aid where it's needed most. In this study, **graduation rate** is the **dependent variable**, and three **independent variables** will be examined:

1. Admission rate
 - a. The amount of selectivity used when accepting applicants. The range is between 0 and 1 where 0 = very selective and 1 = open admission.
2. Cost of Attendance
 - a. Total price for a full-time, in-state undergraduate student.
3. Percentage of Pell Grant Recipients
 - a. Percentage of undergraduates who receive tuition aid with Pell Grants.

Combined, the above listed factors may help explain the difference in completion rates across the colleges in the United States. In this study we explore:

How do admission policy, cost of attendance, and the percentage of Pell Grant recipients relate to graduation rates at four-year U.S. colleges?

Data Sources | Data Cleaning

The dataset retrieved to assist with answering the above question is from the U.S. Department of Education. This is a government site; it helps signify the credibility of the dataset. The site receives its information from other credible sites:

- IPEDS (Integrated Postsecondary Education Data System)
- NSLDS (National Student Loan Data System)
- Treasury Department and IRS Data
- FAFSA (Free Application for Federal Student Aid)

The data needed to be filtered as there were thousands of variables and under 20 that had potential interest for this analysis. While first working with the data, the attempt was made to use ADM_RATE as a binary, categorical variable, but this turned into data that was difficult to interpret. For those reasons, there was a revision made that will be reflected in the final analysis to use ADM_RATE_ALL that will include all students and also uses a decimal value to represent the data.

Descriptive Statistics | Data Visualizations

As shown below in Table 1, on average, college institutions in the dataset admitted approximately 71.6% of their applicants, with admission rates ranging from 0 (very selective) to 1 (open admission). The average annual cost of attendance was \$29,298.41, although the variance is large – ranging from \$5,500 to \$86,964. Further, 43% of students

received Pell Grants, indicating that almost half of students needed financial assistance across college institutions. Finally, the average graduation rate is 52%, meaning that just over half of first-time, full-time students completed their degree within six years.

Table 1

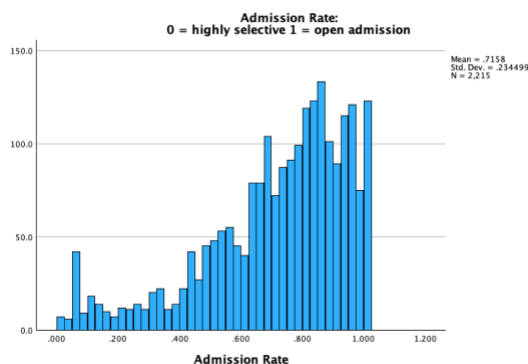
Descriptive statistics: admission rate, cost of attendance, Pell recipients, and graduation rate

	N	Minimum	Maximum	Mean	Std. Deviation
ADM_RATE_ALL	2215	.000	1.000	.71580	.234499
Cost Annually	3316	5500	86964	29298.41	17702.595
Pell Recipients	5671	0	1	.43	.213
Graduation Rate	2280	0	1	.52	.221
Valid N (listwise)	1689				

The histogram shown in Table 2 displays the distribution of admission rates across U.S. colleges in the dataset. Admission rates range from 0 (highly selective) to 1 (open admission). The distribution is right-skewed, showing a majority of college institutions admit between 70% - 100% of applicants. This distribution suggests that most colleges within the dataset have relatively lenient admission policies, making it less common for institutions to have high levels of selectivity regarding admission policies. The average admission rate reflects this at approximately 71.6%.

Table 2

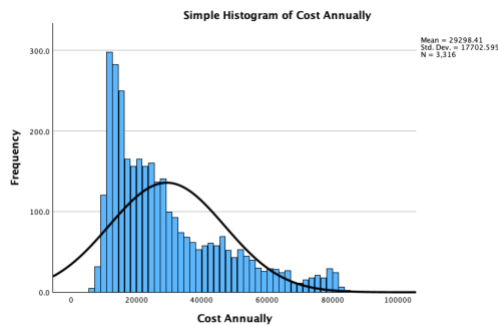
Admission Rate



Moving to look at Table 3 below, the histogram shows the distribution for cost of attendance across college institutions in the dataset. Most college institutions fall within the range of \$10,000 - \$30,000 annually to attend college, with an average annual cost of \$29,298. There are a small number of institutions within the dataset that charge a significantly higher tuition amount, resulting in the long tail shown in the histogram.

Table 3

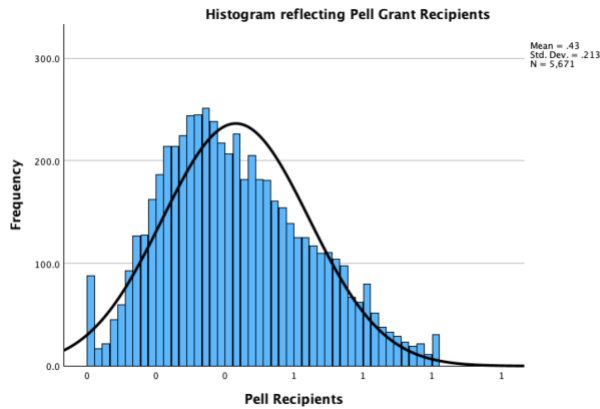
Cost Annually



Most colleges have between 20% and 60% (with an average of 43%) of students receiving Pell Grants, as shown in Table 4 below. Although the distribution is slightly skewed left, it still suggests a moderate to high level of financial assistance among students attending the college institutions in this dataset.

Table 4

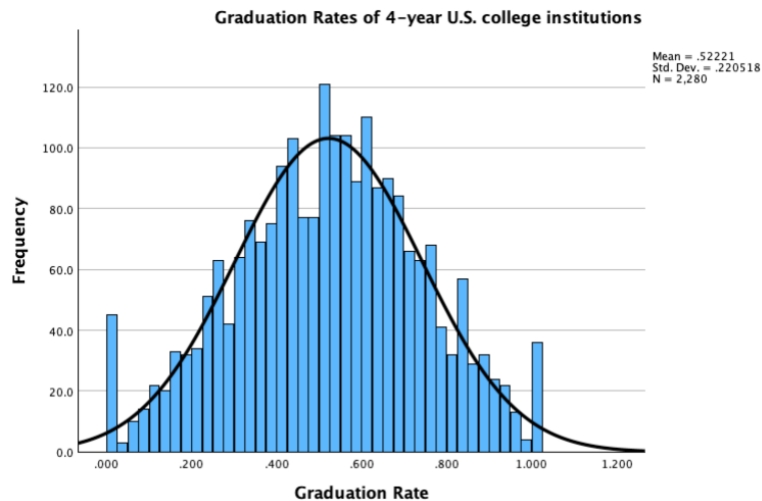
Histogram Reflecting Pell Grant Recipients



When graduation rates are analyzed, approximately 52% of students graduate within six years of enrollment. The distribution is relatively symmetrical and centered around the mean, suggesting that most institutions have graduation rates in the range of 40% and 70%. The normal distribution of this histogram is displayed in Table 5 below.

Table 5

Graduation Rates of 4-year U.S. college institutions



Analytical Model | Data Analysis | Results

A multiple linear regression model was used to examine whether admission rate, cost of attendance, and Pell Grant percentage could predict graduation rates at 4-year college institutions included in this dataset. Overall, the model was statistically significant

based on the resulting F statistic being extremely large and explaining variation in graduation rate ($F = 362.71$) and the p-value being smaller than 0.05 ($p < 0.001$). Further, $R^2 = 0.392$ meaning that 39.2% of the variance in graduation rates is explained by this model.

All three predictor variables have shown to be statistically significant as shown below in Figure 1. Admission rate was negatively related to the graduation rate as shown by $\beta = -0.13$; this means that colleges with more selective admission policies tend to have higher graduation rates. Cost of attendance was positively related to graduation rates ($\beta = 0.34$) suggesting that more costly college institutions tend to have higher graduation rates. Finally, Pell Grant percentages had the strongest negative relationship ($\beta = -0.34$), showing that colleges with more students in need of tuition assistance tend to have lower graduation rates.

Figure 1

Analytical Model and Results

Model Summary				
Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.626 ^a	.392	.391	.155373

a. Predictors: (Constant), Pell Recipients, ADM_RATE_ALL, Cost Annually

ANOVA ^a						
Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	26.269	3	8.756	362.714	<.001 ^b
	Residual	40.677	1685	.024		
	Total	66.946	1688			

a. Dependent Variable: Graduation Rate

b. Predictors: (Constant), Pell Recipients, ADM_RATE_ALL, Cost Annually

Coefficients ^a							
Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.	Collinearity Statistics
		B	Std. Error	Beta			Tolerance VIF
1	(Constant)	.651	.023		27.879	<.001	
	ADM_RATE_ALL	-.119	.019	-.127	-6.243	<.001	.865 1.156
	Cost Annually	3.672E-6	.000	.337	14.917	<.001	.705 1.419
	Pell Recipients	-.418	.027	-.335	-15.691	<.001	.790 1.266

a. Dependent Variable: Graduation Rate

Conclusion

This analysis explored how admission rates, cost of attendance, and the percentage of Pell Grant recipients relate to graduation rates at four-year U.S. college institutions. Using a dataset retrieved from the U.S. Department of Education, descriptive analysis showed that most college institutions have relatively high admission rates and moderate to high levels of financial assistance needed among their students. Graduation rates varied widely across college institutions, with an average of 52% of students completing their degrees within six years (150% of the expected time).

The multiple regression analysis showed that all three independent variables significantly predicted graduation rates. Colleges with lower admission rates (more selective of admission) and higher costs tended to have higher graduation rates. In contrast, colleges with a higher percentage of Pell Grant recipients typically had lower graduation rates. Understanding these significant relationships can help educators and policymakers design more targeted support strategies to improve graduation rates.

References:

National Center for Education Statistics. (2023). *Undergraduate retention and graduation rates*. In *The condition of education 2023*. U.S. Department of Education, Institute of Education Sciences. <https://nces.ed.gov/programs/coe/indicator/ctr>

U.S. Department of Education. (n.d.). *Open admissions*. College Scorecard. <https://collegescorecard.ed.gov/glossary/#open-admissions>

Appendix A

SPSS Syntax Used for Analysis

```

1
2 * Chart Builder.
3 GGRAPH
4 /GRAPHDATASET NAME="graphdataset" VARIABLES=ADM_RATE_ALL MISSING=LISTWISE REPORTMISSING=NO
5 /GRAPHSPEC SOURCE=INLINE.
6 BEGIN GPL
7 GUIDE: axis(dim(1), label("ADM_RATE_ALL"))
8 GUIDE: axis(dim(2), label("Frequency"))
9 GUIDE: text.title(label("Simple Histogram of ADM_RATE_ALL"))
10 ELEMENT: interval(position(summary.count(bin.rect(ADM_RATE_ALL))), shape.interior(shape.square))
11 END GPL.
12
13 * Chart Builder.
14 GGRAPH
15 /GRAPHDATASET NAME="graphdataset" VARIABLES=COSTT4_A MISSING=LISTWISE REPORTMISSING=NO
16 /GRAPHSPEC SOURCE=INLINE.
17 BEGIN GPL
18 GUIDE: axis(dim(1), label("Cost Annually"))
19 GUIDE: axis(dim(2), label("Frequency"))
20 GUIDE: text.title(label("Simple Histogram of Cost Annually"))
21 ELEMENT: interval(position(summary.count(bin.rect(COSTT4_A))), shape.interior(shape.square))
22 ELEMENT: line(position(density.normal(COSTT4_A)))
23 END GPL.
24
25 * Chart Builder.
26 GGRAPH
27 /GRAPHDATASET NAME="graphdataset" VARIABLES=PCTPELL MISSING=LISTWISE REPORTMISSING=NO
28 /GRAPHSPEC SOURCE=INLINE.
29 BEGIN GPL
30 GUIDE: axis(dim(1), label("Pell Recipients"))
31 GUIDE: axis(dim(2), label("Frequency"))
32 GUIDE: text.title(label("Simple Histogram of Pell Recipients"))
33 ELEMENT: interval(position(summary.count(bin.rect(PCTPELL))), shape.interior(shape.square))
34 ELEMENT: line(position(density.normal(PCTPELL)))
35 END GPL.
36
37 * Chart Builder.
38 GGRAPH
39 /GRAPHDATASET NAME="graphdataset" VARIABLES=C150_4 MISSING=LISTWISE REPORTMISSING=NO
40 /GRAPHSPEC SOURCE=INLINE.
41 BEGIN GPL
42 GUIDE: axis(dim(1), label("Graduation Rate"))
43 GUIDE: axis(dim(2), label("Frequency"))
44 GUIDE: text.title(label("Simple Histogram of Graduation Rate"))
45 ELEMENT: interval(position(summary.count(bin.rect(C150_4))), shape.interior(shape.square))
46 ELEMENT: line(position(density.normal(C150_4)))
47 END GPL.
48
49 REGRESSION
50 /MISSING LISTWISE
51 /STATISTICS COEFF OUTS R ANOVA COLLIN TOL
52 /CRITERIA=PIN(.05) POUT(.10) TOLERANCE(.0001)
53 /NOORIGIN
54 /DEPENDENT C150_4
55 /METHOD=ENTER ADM_RATE_ALL COSTT4_A PCTPELL.
56

```